LAMA NACHMAN
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Unseen Complexities of Responsible AI

An eye-opening conversation with Intel's **Lama Nachman**

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When we talk about artificial intelligence, the conversation often gravitates toward its tangible impacts. Yet, lurking beneath these visible advancements are intangible unknowns that most people don't fully grasp. To shed light on these hidden challenges, I interviewed Lama Nachman, Intel Fellow and Director of the Intelligent Systems Lab at Intel. Nachman is at the forefront of AI research and development, steering projects that push the boundaries of what's possible while grappling with the ethical implications of these technologies.

THE INTANGIBLE UNKNOWNS IN RESPONSIBLE AI

"While technical aspects like algorithm development are well understood, the intangible unknowns lie in the intersection of stakeholder needs and the AI lifecycle," Nachman began. She highlighted that these challenges manifest in subtle ways that aren't immediately apparent to most people.

"From algorithmic bias causing invisible but significant harm to certain populations, to the complex balance of automation versus human intervention in the workforce," she explained, "less obvious challenges include building genuine trust beyond technical reliability and the environmental impact of AI systems." One pressing issue is the advent of large language models. "With large language models, it has gotten much harder to test for safety, bias, or toxicity of these systems," Nachman noted. "Our methods must evolve to establish benchmarks and automated testing and evaluation of these systems. In addition, protecting against misuse is much harder given the complexity and generalizability of these models."

APPROACH TO ETHICAL AI

As a pioneer in technology, Intel recognises the ethical implications that come with advancing AI technologies. Nachman emphasised Intel's commitment to responsible AI development. "At Intel, we are fully committed to advancing AI technology in a responsible, ethical, and inclusive manner, with trust serving as the foundation of our AI platforms and solutions," she said.

Intel's approach focuses on ensuring human rights, privacy, security, and inclusivity throughout their AI initiatives. "Our Responsible AI Advisory Council conducts rigorous reviews of AI projects to identify and mitigate potential ethical risks," Nachman explained. "We also invest in research and collaborations to advance privacy, security, and sustainability in AI, and engage in industry forums to promote ethical standards and best practices." Diversity and inclusion are also central to Intel's strategy. "We understand the need for equity, inclusion, and cultural sensitivity in the development and deployment of AI," she stated. "We strive to ensure that the teams working on these technologies are diverse and inclusive."

She highlighted Intel's digital readiness programs as an example. "Through Intel's digital readiness programs, we engage students to drive awareness about responsible AI, AI ethical principles, and methods to develop responsible AI solutions," according to Nachman. "The AI technology domain should be developed and informed by diverse populations, perspectives, voices, and experiences."

OF CHALLENGES AND LESSONS LEARNED

Implementing responsible AI practices comes with its own set of challenges. Nachman was candid about the obstacles Intel has faced. "A key challenge we have as developers of multi-use technologies is anticipating misuse of our technologies and coming up with effective methods to mitigate this misuse," she acknowledged. She pointed out that consistent regulation of use cases is an effective way to address technology misuse. "Ensuring environmental sustainability, developing ethical AI standards, and coordinating across industries and governments are some of the challenges that we as an industry